

BASE GUIDE 3 OF 3

HOW We Execute the Research

Corrected step-by-step experimental protocol

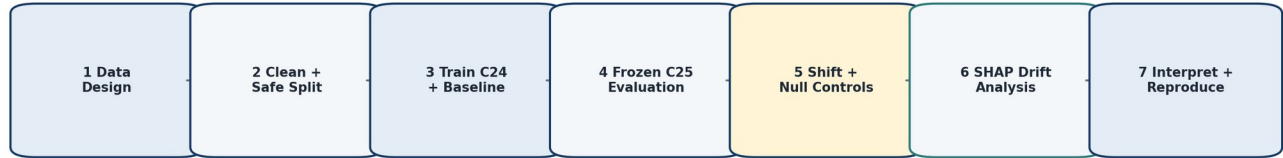
Cross-Cycle Explanation Drift in Solar Flare Forecasting

Version 2.0 | October 2026 | GitHub Base Research Documentation

Status	REVIEWED / BASELINE LOCKED
Research domain	Space Weather Machine Learning Explainable AI

1. Full Pipeline

Seven-Phase Execution Pipeline



Phase gates: each phase has required outputs + an exit criterion before moving forward

Figure 1. Seven research phases. Each phase has a gate before progression.

Locked Research Architecture (v2)

Conceptual workflow - not experimental results

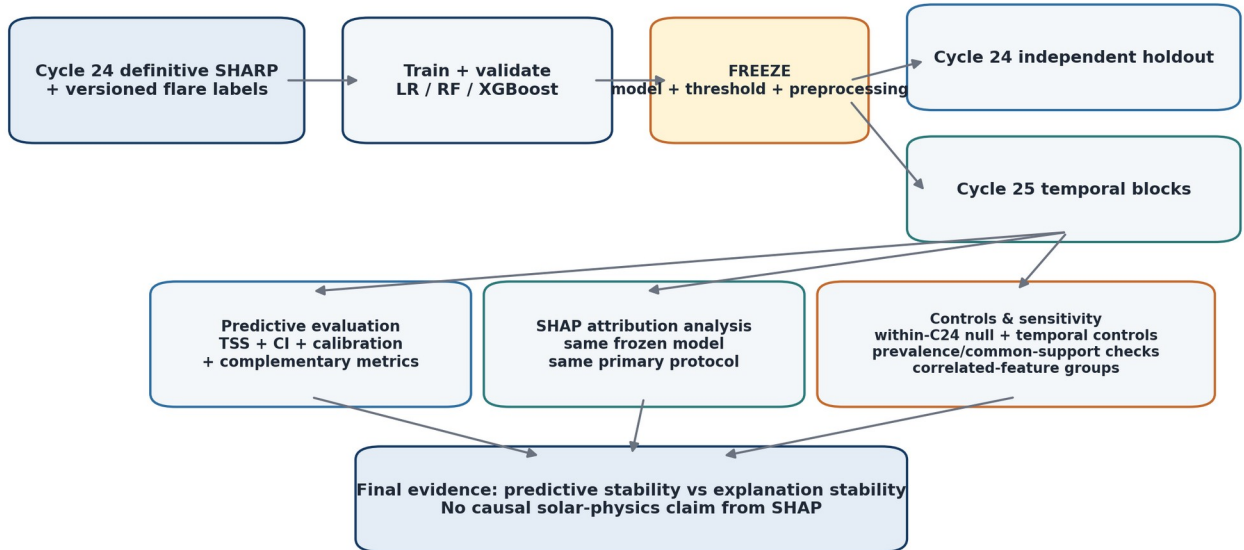


Figure 2. End-to-end model, evaluation, explanation, and control architecture.

2. Phase 1 - Data Collection & Dataset Design

WHAT	WHY	HOW
Build a common Cycle-24/Cycle-25 SHARP + flare-label dataset.	Cross-cycle comparison fails if the two periods are constructed using different definitions.	Lock definitive JSOC SHARP series, query metadata, QUALITY rules, label source, forecast window, sample stride, and event-association rule before modeling.

- Use definitive SHARP, not NRT, for the primary scientific analysis.
- Preferred series for the locked design: hmi.sharp_cea_720s; exact series/version recorded.
- Record the native 12-minute cadence and pre-register the model sampling stride (for example, a coarser snapshot interval) before outcome analysis.
- Primary target: $\geq M$ -class event in the next 24 hours.
- Use one versioned primary GOES/NOAA flare catalog and document how NOAA event IDs are associated with HARP/SHARP records.
- Save raw query metadata, feature dictionary, label specification and data provenance.

PHASE 1 EXIT GATE

The data can be regenerated with the same schema, cadence/stride, quality rules and labels from recorded queries.

3. Phase 2 - Cleaning, Preprocessing & Leakage-Safe Splitting

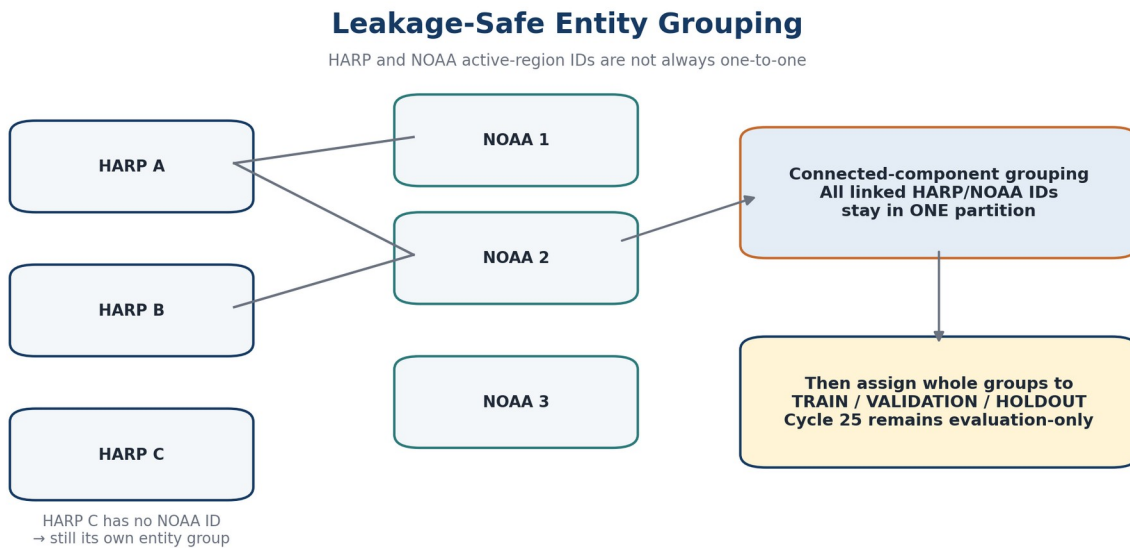


Figure 3. Corrected split unit: connected HARP-NOAA entity group, not blind HARPNUM-only splitting.

- Create connected entity groups from HARPNUM $\leftrightarrow$ NOAA_AR(S) associations; a linked group is indivisible across train/validation/holdout.
- HARPs with no NOAA ID form standalone groups.
- Remove/flag records failing pre-defined QUALITY rules; apply the disk-position/limb rule consistently.
- Fit imputation, scaling, feature selection (if any) and class weighting only on Cycle-24 training data.
- Cycle 25 is completely excluded from model selection and preprocessing fitting.
- Save split manifests with group IDs and audit that no linked entity crosses partitions.

PHASE 2 EXIT GATE

Zero linked HARP/NOAA entity overlap across partitions; every learned transformation is fitted on Cycle-24 training only.

4. Phase 3 - Cycle 24 Training & Historical Baseline

Model	Role	Primary rule
Logistic Regression	Transparent linear baseline	Scaling/preprocessing fitted only on C24 train.
Random Forest	Non-linear ensemble baseline	Hyperparameters selected only on C24 validation.
XGBoost	Strong tabular learner	Hyperparameters and probability threshold locked before C25.

- Train and tune using Cycle-24 train/validation groups only.
- Choose classification threshold using Cycle-24 validation only.
- Freeze model parameters, preprocessing, feature order, and threshold.
- Evaluate once on the independent Cycle-24 holdout.
- Report TSS + entity-level bootstrap CI, plus HSS/F1, PR-AUC, Brier score and calibration as appropriate.
- Pre-register the rule for calling predictive performance "stable" versus "changed" before final Cycle-25 outcome review.

5. Phase 4 - Frozen-Model Cycle 25 Evaluation

- Apply exactly the frozen Cycle-24 pipeline to pre-defined Cycle-25 temporal blocks.
- No Cycle-25 retraining, fine-tuning, threshold tuning, feature selection, or hyperparameter selection.
- Compute the same predictive metrics and uncertainty for every block.
- Treat Cycle 25 as a temporal/domain-shift transfer test; reserve "OOD" terminology for cases supported by feature-support diagnostics.

PHASE 4 EXIT GATE

Every Cycle-25 prediction is produced by artifacts frozen before Cycle-25 evaluation, and block-wise predictive differences have uncertainty estimates.

6. Phase 5 - Distribution Shift & Null Controls

Evidence Ladder Before Declaring Explanation Drift

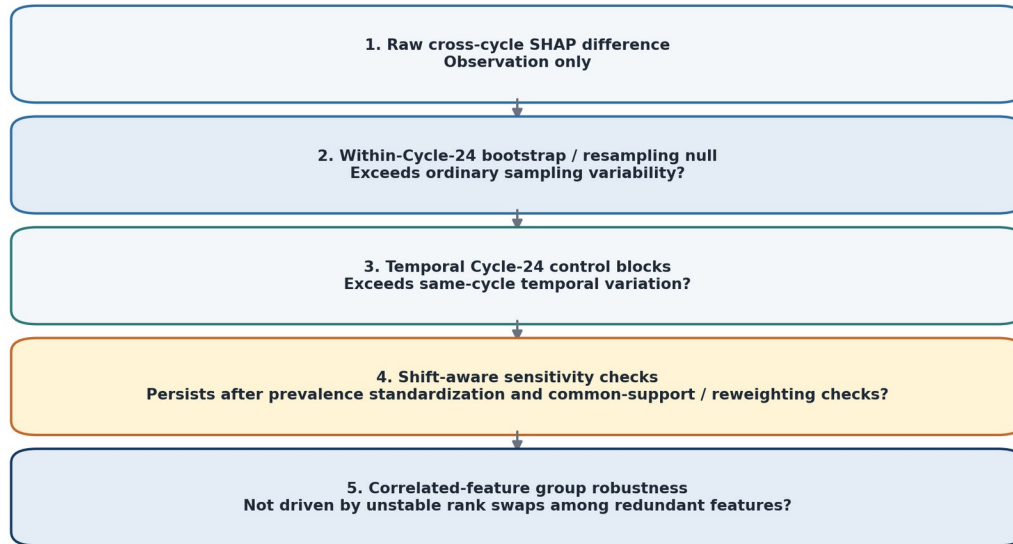


Figure 4. Controls required before an explanation-drift claim.

Control	Implementation
Feature shift	Compare marginal distributions, selected joint structure, missingness, disk position and region properties.
Prevalence shift	Compare $\geq M$ positive rates by cycle and temporal block.
Bootstrap null	Repeated entity-group resampling within Cycle 24 at matched sample sizes.
Temporal null	Compare separated Cycle-24 blocks/phases under the same SHAP protocol.
Class-conditional sensitivity	Compare SHAP within positive and negative labels separately.
Prevalence-standardized sensitivity	Recompute aggregate attribution summaries under a common class mix.
Common-support/reweighting	Where feasible, compare matched or weighted samples with overlapping feature support.
Label-source sensitivity	If feasible, repeat key labels/results with a secondary catalog definition.

7. Phase 6 - SHAP Explanation-Drift Analysis

Element	Protocol
Model	Always compare the SAME frozen model across Cycle-24 holdout and Cycle-25 blocks.
Background/reference	Fixed primary background sampled from Cycle-24 training data; alternatives only as sensitivity checks.
Explainer	Model-appropriate SHAP explainer with package/version/settings recorded.
Global magnitude	Mean $ \text{SHAP} $ by feature and grouped features.
Direction	Sign/direction consistency and response patterns where meaningful.
Ranking	Spearman + Kendall + top-k overlap; never rely on rank alone.
Distribution	Attribution-distribution distance / two-sample statistic selected before final testing.
Null inference	Empirical p-value/threshold from Cycle-24 null controls; FDR for multiple feature-wise tests.
Correlation robustness	Repeat at grouped-feature level for strongly correlated SHARP variables.

DRIFT RULE

A cross-cycle SHAP difference is not called meaningful merely because a rank changed. It must exceed the relevant within-Cycle-24 null and remain qualitatively/statistically robust under the pre-specified sensitivity analyses.

8. Phase 7 - Interpretation & Reproducibility

Pre-Registered Interpretation Matrix

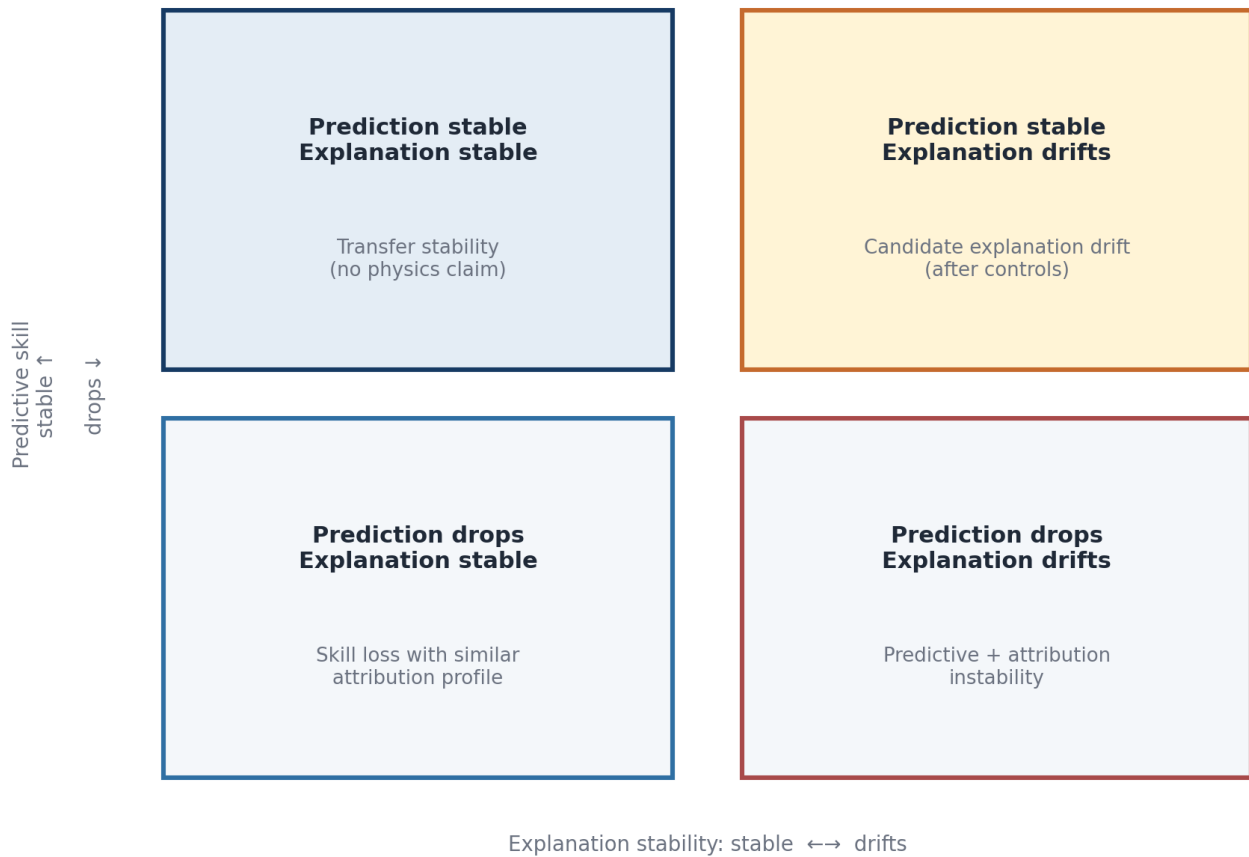


Figure 5. Final interpretation matrix used after all controls.

- Use only the pre-defined result categories; do not invent a stronger narrative after seeing results.
- Separate model-behavior evidence from statements about solar physics.
- Save every query, entity split, configuration, seed, environment, model artifact, prediction file, SHAP output and figure script.
- Re-run the literature/novelty audit immediately before submission.
- Tag the final GitHub release and add CITATION.cff / license information.

9. Phase Gates and Required Artifacts

Phase	Must exist before moving on
1	Versioned queries, product/quality specification, feature dictionary, label rule, stride/cadence rule.
2	Entity-group manifest, QC report, fitted C24 preprocessing pipeline.
3	Frozen models + thresholds + C24 holdout metrics/CI.

Phase	Must exist before moving on
4	C25 block predictions + predictive comparison.
5	Shift report + bootstrap null + temporal null + sensitivity datasets/weights.
6	SHAP outputs + drift statistics + null/FDR results + grouped-feature analysis.
7	Final result matrix + reproducible release package + risk register.

10. GitHub-Ready Directory Structure

```
docs/
  master/
    base_guides/
configs/
data_manifests/
  entity_splits/
src/
  data/
  models/
  evaluation/
  explainability/
experiments/
results/
figures/
tests/
README.md
requirements.txt or environment.yml
CITATION.cff
```

11. Shortest HOW

EXECUTION SUMMARY

Lock data rules -> collect definitive SHARP + one flare catalog -> build leakage-safe connected entity groups -> fit preprocessing and train on Cycle 24 only -> freeze everything -> evaluate Cycle-24 holdout -> evaluate Cycle-25 blocks -> audit distribution shift -> build sampling + temporal Cycle-24 SHAP nulls -> run shift-aware sensitivity analyses -> compare SHAP -> test correlated-feature robustness -> apply pre-registered interpretation matrix -> package reproducible artifacts.

Reference Basis

These references support the current design. They are not the final literature review; novelty-sensitive claims must be re-checked before submission.

[1] Bobra et al. (2014), SHARP data products and HARP/NOAA mapping, Solar Physics. [Source](#)

[2] Hoeksema et al. (2014), HMI vector magnetic-field pipeline and definitive SHARP products, Solar Physics. [Source](#)

[3] Chen et al. (2019), SHARP flare precursors and correlated-feature instability, Space Weather. [Source](#)

[4] Goodwin et al. (2024), solar-cycle and training-window effects in simulated real-time flare forecasting, ApJ. [Source](#)

[5] Hu et al. (2025), data-source quality issues and Cycle-24-to-25 evaluation, arXiv:2512.13417. [Source](#)

[6] Wu et al. (2026), interpretable major-flare forecasting with SHAP, MNRAS. [Source](#)

[7] Li et al. (2026), explainable operational flare forecasting with SHAP, Space Weather. [Source](#)

[8] Roy et al. (2026), SuryaBench, Scientific Data. [Source](#)

[9] Liu et al. (2026), FlareDB for Solar Cycles 24 and 25, Scientific Data. [Source](#)